

Wellbotics
Queen Game
Sample Lesson

Sample Lesson Plan: Sensing and Responding to Touch in Queen

Target Audience: Learners aged 5 - 8 years

Overall Learning Goals:

Programming basics (sequencing and debugging) and Social-Emotional Learning (SEL) (self-awareness, persistence, and mindfulness practice).

Specific Lesson Objectives:

- Students will understand the concept of input and output by observing how Bee-Bots respond to programmed commands.
- Students will begin to recognize and connect touch responses to emotions in themselves and through the robot's "reactions".
- Students will practice sequencing and debugging simple programs for their Bee-Bot.
- Students will engage in collaborative problem-solving and persistence when facing unexpected challenges.

Duration: 45 minutes (following Wellbotics' typical structure: ~10 min Warm Up, ~25 min Work, ~10 min Wrap Up)

Materials:

- Bee-Bots (programmable robots)
- "Garden" game board tiles (movable if possible, or pre-arranged)
- "Collect Pollen" cards (with bright illustrations of happy flowers/pollen)

- "Obstacle" cards (with icons like a fallen branch, strong wind, tangled vines, and prompts for re-programming)
- "Dance" cards (with dynamic bee/Bee-Bot illustrations and emoticons)
- Small physical "pollen" tokens (e.g., pompoms, colorful beads, small blocks)
- Emotion cards/visuals (happy, sad, frustrated, calm)
- Facilitator's costume/props for the "Queen Bee" to visibly represent hive health.

1. Warm Up: Move Like a Bee-Bot, Feel Like a Bee! (~10 minutes)

- Activity:
 - "Move Like a Robot!" (Gross Motor Warm-Up): The "Queen Bee" facilitator leads students in a gross motor activity that simulates step-by-step robot movement. For example, the facilitator might call out, "Take two steps forward!" (students take two steps), "Turn left!" (students turn their bodies left). This helps students understand sequential instructions.
 - "Find Feelings!": Show various emotion cards (e.g., happy, sad, frustrated). Ask students to identify the emotion and mimic the expression or a body pose for it. The facilitator can then ask: "How might a Bee-Bot show it's 'happy' or 'stuck' through its movements?" This introduces the idea of robots showing "feelings" through their actions.
- SEL Connection: Introduces self-awareness by recognizing and naming basic emotions.

2. Work: Bee-Bot's Touch Challenge (~25 minutes)

- Activity:
 - Introduction: The facilitator introduces the game board as the "garden" and explains that Bee-Bots (students' "worker bees") need to collect pollen to keep the hive healthy.
 - Understanding Input/Output (Sensing Touch): Explain that Bee-Bots respond to touch (their buttons) and that different "touches" (cards) will make them do different things. Discuss how humans sense touch and react.
 - Pollen Collection: Students, working in small teams, take turns drawing a "Collect Pollen" card from a designated spot. The card will show a type of pollen to collect.
 - Programming the Path: Teams plan and program their Bee-Bot's path on the game board using directional commands (forward, backward, turn left, turn right) to reach a "pollen" tile. Emphasize chaining 2-3 steps logically.

- Encountering Obstacles (Debugging & Persistence): If a Bee-Bot lands on a tile with an "Obstacle" card, the team draws it. Examples: "Windy Path! Move back 1 space," or "Tangled Vines! Use 1 extra programming card this turn."
 - The facilitator guides teams to "debug" their program when it doesn't work as planned. Encourage them to discuss as a team what went wrong and how to fix it.
 - Use prompts like: "Bumpy Road! It's tough, but don't give up!" to foster persistence in the face of setbacks.
- Dance Moment: After successfully collecting a pollen token (or encountering a specific obstacle), the team might draw a "Dance" card. The facilitator leads a simple, expressive movement related to the card, such as the "Happy Hive Dance!" or "Frustration Station! Take a deep breath. Can you re-program?" (connecting movement to emotions and mindfulness).
- SEL Connection: Promotes persistence and problem-solving. Encourages self-management by working through frustration. Fosters collaboration and communication as teams cooperate on programming and debugging. Introduces mindfulness through guided movement and reflection on emotions.

3. Wrap Up & Reflect: Hive Health Check (~10 minutes)

- Activity:
 - Assess Hive Health: The "Queen Bee" visually updates the hive's health status, showing how the collective efforts of the "worker bees" (students) impact the hive. This can be done through a costume element or prop.
 - "Watch & Share!": Ask students to reflect on their experience. "How did you feel when your Bee-Bot successfully collected pollen? How about when it hit an obstacle? What did you do when you felt frustrated?"
 - Connection to Self: Discuss how programming the robots to sense and respond to "touch" can help them think about their own bodies sensing information and how they respond emotionally.
 - "High-Five & Try Again!": Reinforce that learning from mistakes and trying again is an important part of both coding and life, celebrating their efforts in debugging and teamwork.

- SEL Connection: Encourages deeper self-awareness by connecting physical and emotional responses. Reinforces social awareness and relationship skills by reflecting on collective well-being and the impact of individual contributions on the team.

Assessment Indicators (Informal Observation & Discussion):

- SEL:
 - Can students recognize and name basic emotions as they discuss their feelings during the game (e.g., happy when successful, frustrated when debugging)?.
 - Do students demonstrate persistence by re-attempting a program after an error or obstacle?.
 - Do students show signs of collaboration and communication with their peers (e.g., discussing solutions, sharing the Bee-Bot)?.
- Computer Science/Programming:
 - Can students accurately program a sequence of 2-3 steps for the Bee-Bot?.
 - Can students identify simple errors in their program (debugging) and attempt to correct them?.
 - Do students demonstrate an understanding of input/output by describing how a sensor (like the obstacle card) triggers a robot's response?.

This lesson aims to provide a hands-on, interactive, and multidisciplinary experience, aligning with Wellbotics' mission to combine computer science and digital fluency with social and emotional learning, fostering a dynamic environment where students can explore the intricate connections between robot actions and human behaviors and emotions. The activities are designed to be developmentally appropriate for grades K-2.